

Final Presentation: A Review of Distributed Energy Resources and Modern Electrical Grid Optimization Methods

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Introduction to Distributed Energy Resources and Modern Electrical Grid Optimization Methods

Introduction

- ❖ **Distributed Energy Resources (DERs)** are small scale energy generation and storage technologies located near the point of use.
- ❖ **Reductions of cost** for common DERs, like photovoltaic (**PV**) cells and battery energy storage systems (**BESS**), has opened these technologies to the wider market in recent years.
- ❖ DERs introduce unique challenges to the modern grid by increasing the number of generation assets, **introducing fast-dynamic** devices like inverters, and **reducing predictability** in energy production capacity.

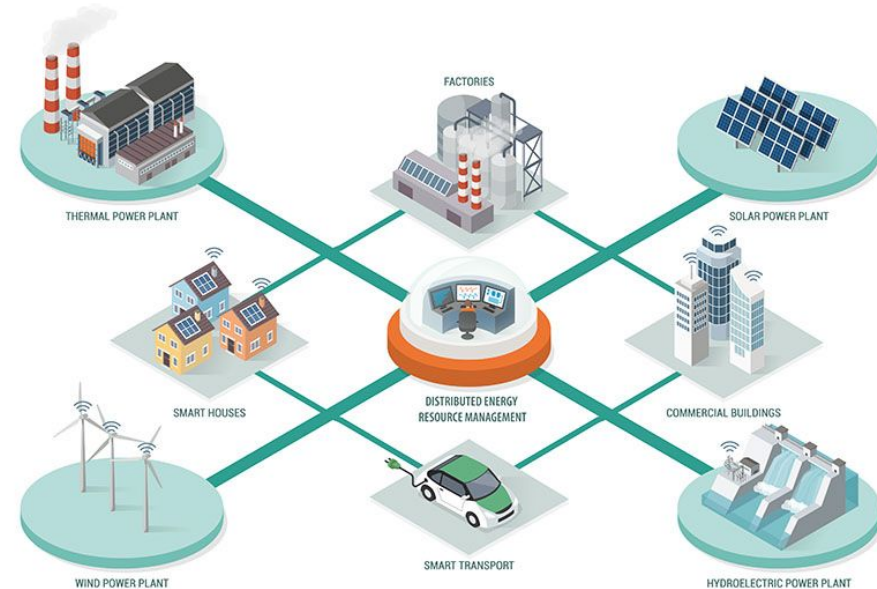


Figure: Conventional grid layout with DERs, featuring larger thermal facilities, renewable deployments, and residential systems.

Introduction: Motivation

- ❖ The nature of DERs necessitates a **shift in optimization paradigms** to **ensure continued reliability** and **functionality of the grid** as they take up a larger share of generation and dispatch.
- ❖ **Model assumptions, objective function design, and constraint structures** can dramatically influence optimization solutions and their applicability.



Figure: On-campus testbed DERConnect switchgear room used to create unique on/off grid testing environments

Introduction: Nomenclature

- ❖ **P:** Represents the overarching category the variable is assigned to. (i.e. P is used for Electrical Power in kW.)
- ❖ **INDEX:** Indicates any indexing like nodal value, time, decision step, or specific generator in a system.
- ❖ **DESC_1:** Used to note whether the valuation is an input, output, or forecast/ prediction value.
- ❖ **DESC_2:** Used for any other identifying information, like generator or system variety. (i.e. BESS, Grid, or Wind Turbine)

$$P^{desc_1, desc_2}_{index}$$

Figure: General form used for uniform nomenclature.

Relevance of Contributions

Relevance of Contributions

Subsection	Relevance of Sources in this Section
PV + BESS Systems	❖ The fast dynamics of PV systems can quickly change output, these works demonstrate how convex optimization of PV + BESS systems enables efficient operations that minimize losses and smooth load profiles, produces scalable scheduling and sizing plans, and optimizes placement for some of the most common carbon free DERs .
Microgrids	❖ Convex and Mixed integer methods are shown to assist in real time dispatch, demand response , and long term DER sizing within microgrid environments with proven convergence, reductions in costs , and validated performance in some cases.
Distributed Controls	❖ Showcases that convex optimization can enable fully distributed, real time DER coordination by converting non convex constraints to achieve scalable and near optimal energy system management without a centralized control system .

Technical Description of Contributions and Advantages

Technical Description: PV + BESS

1. Nottrott et al. (LP)

$$\min \{f(P_n^{f,l}, P_n^o)\} = \min \sum_{k=1}^N \left(P_k^{f,l} - P_k^{o,PV+BESS} \right) \Delta t, \\ \text{while } P_k^{f,l} > 0 \text{ and } P_k^{f,l} > P_k^{o,PV} \quad (1)$$

2. Li et al. (SOCP)

$$\min \quad f_4 = \sum_t^{\mathcal{H}} \sum_b^{\mathcal{B}} u_{b,t} \quad (2) \\ \text{s.t.} \quad \left\| \begin{array}{c} 2\sqrt{c_{2b}} P_{b,t}^{BUS} \\ 1 + c_{1b} P_{b,t}^{BUS} + c_{1,t} p_t^{Grid} - u_{b,t} + c_{0b} \end{array} \right\|_2 \\ \leq 1 - c_{1b} P_{Gb,t} - c_{1,t} p_{Grid,t} + u_{b,t} - c_{0b} \quad (3)$$

3. Osorio et al. (SOCP)

$$\min \text{Re} \left(\sum_{h \in \mathcal{L}} \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} (\|V_{i,h}\|^2 - W_{a,b,h}) y_{a,b}^* \right), \quad (4)$$

Technical Description: PV + BESS

Author	Nottrott et al.	Li et al.	Osorio et al.
Objective	Consumer electricity cost	Find optimal DES allocation and operation	Distribution network power loss
Decision Variables	Battery power output	DES size, DES location, DES dispatch	DER location, DER capacity, BESS charge/discharge schedule
Optimization Type	Linear Programming (LP)	Second-order cone programming (SOCP)	Second-order cone programming (SOCP)
Constraints	Power output balance, energy conservation, BESS power limits, BESS charge/discharge limits	PV availability, voltage/line limits, SoC dynamics	Power balance, SOC, ramping, solar resources, inverter capacity
Applied Problem	Behind-the-meter system	Southwestern US	San Andres distribution network
Limitations	No Uncertainty Estimate	Limited Variety of Generator Mix	Relaxation Exactness Unverified

TABLE I
COMPARISON OF OPTIMIZATION APPROACHES FOR ENERGY SYSTEM PLANNING IN PV+BESS SYSTEMS

Technical Description: Microgrids

1. Yuan et al. (Co-Optimization)

$$FS = \gamma^{PV} \cdot \sum P_{nom}^{PV} + \gamma^{WT} \cdot \sum P_{nom}^{WT} + \gamma^{Biomass} \cdot P_{nom}^{Biomass} + \gamma^{NG} \cdot \sum P_{thr}^{NG} \quad (5)$$

$$ER = \alpha^{PV} \cdot \sum P_{nom}^{PV} + \alpha^{WT} \cdot \sum P_{nom}^{WT} + \alpha^{NG} \cdot \sum P_{thr}^{NG} + \alpha^{Biomass} \cdot P_{nom}^{Biomass} \quad (6)$$

$$SES = \beta^{PV} \cdot \sum P_{nom}^{PV} + \beta^{WT} \cdot \sum P_{nom}^{WT} + (\beta_e + \beta_{th}) \cdot \left(\sum P_{thr}^{NG} + P_{nom}^{Biomass} \right) \quad (7)$$

2. Cai et al. (Convex MPC)

$$\min \left\{ \sum_{n=1}^{\mathcal{H}} \{P_n^{grid} \cdot c_n^e + \sum_{j=1}^3 (Q_{j,n}^{ht} \cdot c^{gas})\} + \sum_{l=1}^{\mathcal{L}} \{\max(\max_{n \in P_l} (P_n^{grid}) - P_{l,k}^{thresh}, 0) \cdot c_l^{DC}\} \right\} \quad (8)$$

3. Elkazaz et al. (Hybrid Microgrid)

$$C^{OP, MG} = C^{i, MG} - C^{o, MG} + C^{OP, BESS} \quad (9)$$

$$\min_U \quad \frac{1}{2} U^T H_i U + G_i^T U \quad (10)$$

4. Liu et al. (Distributed Fixed-Time)

$$\min \sum_{i=1} f_i(P_i) = \left[\sum_{i=1}^{n_1} C_i^{CG} + \sum_{i=1}^{n_3} C_i^{Bess} - \sum_{i=1}^{n_4} C_i^U \right] \quad (11)$$

Technical Description: Microgrids

Author	Yuan et al.	Cai et al.	Elkazaz et al.	Liu et al.
Objective	Multi-objective: fuel savings, emissions, policy compliance	Operational cost minimization	Daily operating cost + degradation cost	Maximize social welfare (utility of consumption minus cost of generation)
Decision Variables	DER capacities, load allocation	Power flows, SOC, thermal control	Dispatch, SOC, grid exchange	Local generation, load consumption, storage dispatch
Optimization Type	KKT-based, multi-stage nonlinear	Convex (MPC)	LP + QP (MPC)	Lagrangian Duality
Constraints	Power balance, policy mandates, DER limits, KKT conditions	Power balance, SOC, PV limits, thermal dynamics	Power balance, SOC, ramping, grid limits, degradation	Power balance, device limits, consensus dynamics
Applied Problem	Ohio microgrid	3-building microgrid	Hybrid PV-wind microgrid	6-bus and 30-bus microgrid systems
Limitations	No Network Constraints or BESS Cycling Constraints	No Network Constraints	Single Model Day Validation Only	No AC Power Flow or BESS degradation protection

TABLE II
COMPARISON OF OPTIMIZATION APPROACHES FOR ENERGY SYSTEM PLANNING IN MICROGRIDS

Technical Description: Distributed Control

1. Lindberg et al. (MILP ZEB)

$$\begin{aligned} \min \pi = & \sum_{i \in I} (C_i^{\text{totspec}} x_i + C_i^{\text{totfixed}}) \\ & + \sum_{h=1}^H \frac{1}{(1+r)^{(h-1) \cdot N(h)}} \cdot \sum_{n=1}^N \frac{C_h^{\text{totrun}}}{(1+r)^n} [EUR] \end{aligned} \quad (12)$$

2. Muñoz-Salcedo et al. (MIDCP)

$$\min_{z,x} \quad \sum_{i \in \mathcal{G}} C_i^{\text{inv}} z_i + \sum_{n \in \mathcal{H}} \sum_{i \in \mathcal{G}} C_i^{\text{op}} x_{i,n} + \sum_{i \in \mathcal{G}} C_i^{\text{rep}} z_i + \sum_{n \in \mathcal{H}} c^{\text{CO}_2} EM \quad (13)$$

Technical Description: Distributed Control

Author	Muñoz-Salcedo et al.	Lindberg et al.
Objective	Lifecycle cost minimization	Lifecycle ZEB cost minimization
Decision Variables	Integer tech selection, continuous sizing	Binary tech selection, sizing
Optimization Type	MIDCP	MILP
Constraints	Energy balance, SOC, emissions, integer limits	ZEB constraints, building physics
Applied Problem	Madrid / Ecuador	ZEB design
Limitations	Restrictive Convexity Assumptions	Low Model Resolution

TABLE III
COMPARISON OF OPTIMIZATION FRAMEWORKS FOR DISTRIBUTED CONTROL ALGORITHMS

Overall Comparisons

1 PV & BESS

1.1 Optimizes a specific DER technology

1.2 Fixated on battery dispatch, location, size and efficiency

2 Microgrids

2.1 Optimization within a single microgrid

2.2 Considers multiple DER technologies, creates much more dynamic problems

3 Distributed Controls

3.1 Optimizing control systems for multiple controllers

3.2 Heavy emphasis on decision making based on hyper-local information

Limitations

Limitations

- ❖ **Model reductions and relaxations**, while essential to convex optimization, often remove vital components of power systems that play a **key role in performance**. (AC power flow, line losses, etc.)
- ❖ **Removal of performance degradation metrics** lead to unrealistic system performances. (BESS charge/ discharge)
- ❖ Speed of Solution formulation and **applicability of prediction horizons**.



Figure: BESS setup at UCSD before utilization at an Arizona utility.

Unanswered Questions & Extensions of Research

Unanswered Questions / Expansion

1 PV & BESS

1.1 Li et al - Integration of Other Variable Renewable Energy systems

1.2 Wind Energy

- Time Scale
- Georgiadis et al - Forecasting

2 Microgrids

2.1 Yuan et al - Additional Objective Function of Power Flow

2.2 Elkazaz et al - MATLAB simulations robustness to real world disturbances

3 Distributed Controls

3.1 Optimization of a neighborhood of individual controllers

3.2 Improved communication between nodes, but mindful of privacy

Questions?

Thank you!

Appendix

Appendix: Relevance of Contributions: PV + BESS Systems

Energy dispatch schedule optimization and cost benefit analysis for grid-connected, photovoltaic-battery storage systems by Nottrott et al.	❖ Model: Linear Programming (LP) to model an idealized PV+ system (PV array, LI battery, connected to utility grid) and ❖ Goal: determine the optimal dispatch schedule for BESS to achieve load peak shaving and reduce utility cost via day ahead PV power output and load forecasts for optimal charge/discharge of BESS. Compares to nonconvex dispatch schedules.
Convex Optimization for DES Planning and Operation in Radial Distribution Systems With High Penetration of Photovoltaic Resources by Li et al.	❖ Model: Two Stage Optimization Method that first determined the appropriate sizing of the DES based on hourly time resolution and determined optimal placement in the power system based on minutely time resolution. ❖ Goal: Grids with high PV penetration need to be coupled with robust distributed energy storage (DES) to alleviate the effects of solar generation on the grid.
Optimization of Distributed Energy Resources in Distribution Networks: Applications of Convex Optimal Power Flow Formulations in Distribution Networks by Osorio et al.	❖ Model: Convex formulations of the optimal power flow (OPF) problem for DER (PV & BESS) installations in San Andres distributed network. ❖ Goal: Minimize power losses. over 4 case studies with different formulations: convex power flow analysis, convex optimal power flow to locate and size dispatchable and non dispatchable DER (PV) with demand and irradiation profiles, and the inclusion of aggregated and distributed PV and BESS systems. ❖ Convex formulations help substantially to reduce computation times, especially if scalability is to be considered, and made PV+BESS DERs competitive.

Appendix: Relevance of Contributions: Microgrids

Energy management system for hybrid PV-wind-battery microgrid using convex programming, model predictive and rolling horizon predictive control with experimental validation by Elkazaz et al.	❖ Model: Experimental hardware testing for convex optimization models. ❖ Goal: Seeking to address how problem formulation affects real time energy management via the operational outcomes of the models presented. ❖ Applies theory to a real life electrical model to test how quickly a microgrid can respond to rapidly changing grid conditions.
Mixed-integer disciplined convex programming approach applied to the optimal energy supply of near-zero energy buildings by Munoz-Salcedo et al.	❖ Model: Investigates the method of mixed integer linear programming in relation to the integration of VRE in nearly zero energy buildings (nZEB). ❖ Goal: Minimize the global cost of building nZEBs to size the long term optimal energy supply.
Co-Optimization Scheme for Distributed Energy Resource Planning in Community Microgrids by Yuan et al.	❖ Goal: Investigate how DER sizing and challenges with planning under regulatory constraints ties together with optimization problem formulation to affect the operational and economic outcomes of DER coordination efforts.
Distributed Fixed-Time Optimal Resource Management for Microgrids by L. Liu et al.	❖ Model: Consensus-gradient algorithm that has a fixed, bounded time proof to convergence of an optimal solution. ❖ Goal: Integration of demand response and economic dispatch to include flexible loads as active market participants. Includes transmission losses and varying communication topologies.

Appendix: Relevance of Contributions: Distributed Controls

Distributed Optimal Coordination for Distributed Energy Resources in Power Systems by D. Wu et al.	❖ Model: Conversion of nonconvex storage constraints to convex equivalents to optimize DER dominant grids. ❖ Goal: Coordination of DERs in real time without a central control system. Solving scalability and communication issues that come with high deployment of DERs with significant capacity for generation and storage.
A Bandit Online Convex Optimization Approach To Distributed Energy Management In Networked Systems by I. Tsetis et al.	❖ Model: Uses convex optimization to model energy sharing between interconnected DER nodes. ❖ Goal: Minimize wasted energy using a distributed-decision making policy to solve the equation for a bounded regret and constraint violations.

Technical Description: Foundational Paper

Zero Duality Gap in Optimal Power Flow Problem by Lavaei et al.

- ❖ **Applied Scope:** Optimal Power Flow (OPF)
- ❖ **Type of Optimization Problem:** Duality, Semidefinite programming (SDP)
- ❖ **Objective Function Used:**

$$h(x, r) := \sum_{k \in \mathcal{N}} \left\{ \underline{\lambda}_k P_k^{\min} - \bar{\lambda}_k P_k^{\max} + \lambda_k P_{D_k} + \underline{\gamma}_k Q_k^{\min} - \bar{\gamma}_k Q_k^{\max} + \gamma_k Q_{D_k} + \underline{\mu}_k (V_k^{\min})^2 - \bar{\mu}_k (V_k^{\max})^2 \right\} \\ + \sum_{k \in \mathcal{G}} (c_{k0} - r_k^2) - \sum_{(l,m) \in \mathcal{L}} \left\{ \lambda_{lm} P_{lm}^{\max} + \mu_{lm} (\Delta V_{lm}^{\max})^2 + (S_{lm}^{\max})^2 r_{lm}^1 + r_{lm}^4 + r_{lm}^6 \right\}$$

Where $P_{D_k} + Q_{D_k}$ is the **given apparent power** of the load connected to the bus k , $k \in \mathcal{N} = \{1, \dots, n\}$, $P_{G_k} + Q_{G_k}$ is the **apparent power** of the generators connect the bus k , V_k is the **complex voltage** at bus k , P_{lm} is the **active power transferred** from the bus k to the through the line $(l, m) \in \mathcal{L} \subseteq \mathcal{N} \times \mathcal{N}$, S_{lm} **apparent power transferred** from bus k through line (l, m) to the network,

Technical Description: Foundational Paper (Continued)

Zero Duality Gap in Optimal Power Flow Problem

- ❖ **Decision Variables:** maximizing $\mathbf{x}$ and $\mathbf{r}$, $\mathbf{x} \geq 0$
- ❖ **Results:** Showed how an NP-hard problem could be relaxed to a SDP problem that can not only be approximated, but solved exactly using zero gap duality.
- ❖ **Comparisons:** All other papers

Technical Description Ex 1

Energy dispatch schedule optimization and cost benefit analysis for grid-connected, photovoltaic-battery storage systems by Nottrott et al.

- ❖ **Applied Scope:** Behind-the-meter PV-BESS system
- ❖ **Type of Optimization Problem:** Linear Programming
- ❖ **Objective Function Used:**

$$\min \{ f(P_{lf}^n, P_o^n) = \sum_{k=1}^N (P_{lf}^k - P_o^k) \Delta t, \text{ while } P_{lf}^k > 0 \text{ and } P_{lf}^k > P_{pf}^k \}$$

where P_{lf} is the forecasted power load, P_o is the combined PV and BESS power output, and P_{pf} is the forecasted PV power output

- ❖ **Decision Variables:** P_s^n which is equal to $P_o - P_{pf}$
- ❖ **Results:** In optimally sized PV+BESS systems, the LP battery dispatch reduced battery cycling in comparison to baseline methods (charge during off-peak, discharge during on-peak hours)
- ❖ **Comparisons:** Convex Optimization for DES Planning and Operation in Radial Distribution Systems With High Penetration of Photovoltaic Resources [Li et al.]

Technical Description Ex 2

Convex optimization-based control of sustainable communities with on-site photovoltaic (PV) and batteries [Cai et al.]

- ❖ **Applied Scope:** Microgrid small community scale
- ❖ **Type of Optimization Problem:** Model Predictive Control (MPC)
- ❖ **Objective Function Used:**

$$\min \left\{ \sum_{i=k}^{Np+k-1} \{P_g[i] \cdot r_e[i] + \sum_{j=1}^3 (Q_{ht}^j[i] \cdot r_{gas})\} \right. \\ \left. + \sum_{l=1}^{Nd} \{ \max(\max_{i \in P_l} (P_g[i]) - \underline{Pow_{thresh,l}}[k], 0) \cdot r_{DC,l} \} \right\}$$

- ❖ **Decision Variables:** Energy Supply Decisions (P_g, Q_{ht}), Power Storage (P_c, P_d), Power Distribution Decisions ($P_{i,t}$)
- ❖ **Results:** Implementing the MPC resulted in 35% demand cost reduction and 20% total cost reduction for small community case study
- ❖ **Comparisons:** Co-Optimization Scheme for Distributed Energy Resource Planning in Community Microgrids [Yuan et al.], Energy management system for hybrid PV-wind-battery microgrid using convex programming, model predictive and rolling horizon predictive control with experimental validation [Elkazaz et al.]

Cai et al Formulation

$$\min \left\{ \begin{aligned} & \sum_{i=k}^{Np+k-1} \{P_g[i] \cdot r_e[i] + \sum_{j=1}^3 (Q_{ht}^j[i] \cdot r_{gas})\} \\ & + \sum_{l=1}^{Nd} \{ \max(\max_{i \in P_l} (P_g[i]) - \underline{Pow_{thresh,l}}[k], 0) \cdot r_{DC,l} \} \end{aligned} \right\}$$

$$P_g[i] = \max(Pow_{tot}[i] + P_c[i]/\eta_c - P_d[i] \cdot \eta_d - P_{PV,max}[i], 0) \quad (1)$$

$$P_{PV}[i] = Pow_{tot}[i] + P_c[i]/\eta_c - P_d[i] \cdot \eta_d - P_g[i] \quad (2)$$

$$SOC[i+1] = SOC[i] + P_c[i] - P_d[i] \quad (3)$$

$$0 \leq SOC[i+1] \leq SOC_{max} \quad (4)$$

$$0 \leq P_c[i] \leq P_{c,max} \quad (5)$$

$$0 \leq P_d[i] \leq P_{d,max} \quad (6)$$

$$\mathbf{x}^j[i+1] = \mathbf{A}^j \mathbf{x}^j[i] + \mathbf{B}_w^j \mathbf{w}^j[i] + \mathbf{B}_u^j Q_z^j[i] \quad (7)$$

$$Q_z^j[i] = -LR^j[i] \cdot Cap_{rate}(T_{oa}[i]) + Q_{ht}^j[i] + Pow_{fan} + m_{oa}^j[i] C_{va}(T_{oa}[i] - T_{RA}) \quad (8)$$

$$T_{z,lb}[i+1] \leq x_z^j[i+1] = C^j \mathbf{x}^j[i+1] \leq T_{z,ub}[i+1] \quad (9)$$

$$m_{oa,min} \leq m_{oa}^j[i] \leq m_{oa,max} \quad (10)$$

$$0 \leq LR^j[i] \leq 1 \quad (11)$$

$$0 \leq Q_{ht}^j[i] \leq Q_{ht,max} \quad (12)$$

Unanswered Questions / Expansion

- **Common Themes:**

- 1 Improvements on Efficiency

- 2 Application Range

- 3 Other Relaxation Methods

- **Paper Specific Questions:**

- Ex 1 Expand to consider multiple buses, additional cost constraints

- Ex 2 Resulting benefits at larger scales



Figure: Sun shining on bright future for DERs